“How a Particular Protein (Tumor Necrosis Factor) Might Control the Immune Response in Prostate Cancer” (Ida Deichaite, PhD) [#118]

Brad Power and Kristen Warren
October 30, 2024

“Blocking TNF with existing, approved FDA drugs may enhance immunotherapy efficacy in prostate cancer by reprogramming the tumor microenvironment to promote a more robust anti-tumor immune response.” – Ida Deichaite, PhD

Meeting Summary

Advanced prostate cancer patients are always searching for more treatment options, especially since each therapy eventually fails. Immunotherapies for cancer are very attractive since they can result in a more durable response, and the side effects can be less severe. Prostate cancer has been considered a harder cancer type to apply immunotherapies to, yet research continues on new approaches. One of these is using a particular protein, the Tumor Necrosis Factor, to control the immune response. The translational goal is to make prostate cancer respond to immunotherapies.

Ida Deichaite, PhD, Assistant Adjunct Professor, Radiation Medicine and Applied Science, at the University of California, San Diego, is uniquely qualified to describe the state of research into therapies using the Tumor Necrosis Factor to modulate the immune response in prostate cancer. Dr. Dechaite's background is in virology and molecular biology. She is conducting genomics analysis to identify susceptibility targets for aggressive cancers. The overarching goal of her research program is to identify and validate actionable targets and new targetable mechanisms for oncology diagnostic and therapeutic development. Her bench-to-bedside approach to studying real-world genomics paired with clinical data facilitates rapid translation into the clinic and access to patients.

What should you know about this novel immunotherapy approach?

This is a hypothesis, like many, of a possible avenue for controlling cancer, in this case focused on prostate cancer. The hypothesis is that blocking a particular protein (Tumor Necrosis Factor) with existing drugs approved by the FDA that may enhance immunotherapy efficacy in prostate cancer by reprogramming the tumor microenvironment to promote a more robust anti-tumor immune response. It is in a research phase and will have to go through clinical trials, which are being developed, before it will be accessible to patients. The opportunity is to launch an exploratory clinical study to test the hypothesis of using TNF inhibitors in combination with checkpoint inhibitors. The study would involve pre-treating patients with TNF inhibitors and analyzing biomarkers of immunosuppression post-treatment.

How can you learn more about this?

- Contact Ida Deichaite at ideichaite@health.ucsd.edu.

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Meeting Notes

KEYWORDS

TNF inhibitors, prostate cancer, immune response, checkpoint inhibitors, tumor microenvironment, inflammatory response, E-selectin, IL-6, RNA sequencing, clinical trial, immunosuppression, biomarkers, patient data, translational approaches, immunotherapy efficacy.

SPEAKERS

Ida Deichaite (70%), Brian McCloskey (14%), Robb Owen (7%), Brad Power (6%), Rick Davis (2%), Saed Sayad (0%), Alane Watkins (0%)

CHAT CONTRIBUTORS

Robb Owen, Rick Davis, Alane Watkins, David Plunkett, Stephen Barry, Alexander Lalov, Ryan Moon, Chris Apfel

SUMMARY

Dr. Ida Deichaite from UC San Diego presented her research on tumor necrosis factor (TNF) and its role in prostate cancer immune response. She hypothesized that blocking TNF with FDA-approved drugs could enhance immunotherapy efficacy by reprogramming the tumor microenvironment. Her study, involving 49 and 169 prostate cancer patients, showed TNF expression increased with disease progression, while IL-6 and E-selectin levels decreased. The data suggested TNF's immunosuppressive role in prostate cancer. She proposed a clinical trial combining TNF inhibitors with checkpoint inhibitors to test this hypothesis. The discussion also explored potential complementary research with CAR-T cell therapy and the use of TNF inhibitors in combination with androgen ablation blockers.

OUTLINE

Introductions

- Dr. Ida Deichaite of UC San Diego has contributed to the field of radiation medicine and applied science.
- She focused here on “translational” approaches and patient groups, aiming to bridge the gap between research and applications that are accessible to patients.
- Her hypothesis is that blocking the tumor necrosis factor (TNF) with FDA-approved drugs may enhance immunotherapy efficacy in prostate cancer.

Overview of TNF's Role in Inflammation and Immunology

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- Dr. Deichaite explains TNF's multiple roles in inflammatory processes, including initiating and regulating the body's response.
- TNF's involvement in activating endothelial cells and increasing vascular permeability to help immune cells reach affected tissues is discussed.
- The dual function of TNF in inflammation and immunosuppression is highlighted, with a focus on its potential role in prostate cancer.
- She mentions the widespread use of TNF inhibitors in autoimmune diseases and their established safety profile.

Genomic Study and Bioinformatics Analysis

- Dr. Deichaite describes the bioinformatics study of prostate cancer genomics, focusing on RNA sequencing data from patients at the Moores Cancer Center (UCSD).
- The study aimed to understand the inflammatory microenvironment in prostate cancer, comparing localized and metastatic disease.
- TNF expression was consistently observed in prostate cancer, with an increase in expression as disease progresses.
- IL-6 expression was found to regress in metastatic disease, contrary to expectations, leading to further investigation.

Correlation with Checkpoint Inhibitors and Immune Cells

- Dr. Deichaite discusses the correlation between TNF expression and checkpoint inhibitors, using data from a study by Caris involving thousands of patients.
- The study showed a statistically significant effect of TNF expression on checkpoint inhibitor expression, suggesting that the cold tumor microenvironment may not be due to a lack of checkpoint inhibitors.
- The analysis of immune cell subgroups showed an increase in M2 macrophages (a critical part of the tumor microenvironment, regulating a wide range of processes, including cancer cell progression) in later stages of prostate cancer.
- The big opportunity is blocking TNF to enhance checkpoint inhibitor efficacy by reprogramming the tumor microenvironment.

E-Selectin and TNF Expression in Prostate Cancer

- Dr. Deichaite explains the role of E-selectin in endothelial cells and its activation by TNF, highlighting its importance in the tumor vasculature.
- The study found a significant reduction in E-selectin protein expression in tumor tissues compared to normal tissues and BPH.
- Correlation with overall survival data showed that high E-selectin expression was associated with significantly better outcomes.
- The dual high expression of E-selectin and low TNF expression was found to be a significant predictor of overall survival in prostate cancer patients.

Clinical Implications and Future Research

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- The opportunity is to launch an exploratory clinical study to test the hypothesis of using TNF inhibitors in combination with checkpoint inhibitors.
- The study would involve pre-treating patients with TNF inhibitors and analyzing biomarkers of immunosuppression post-treatment.
- Robb Owen shares his experience with using TNF inhibitors and other treatments to manage cancer, emphasizing the importance of neurotransmitters and diet in cancer treatment.
- Brad Power raises questions about the broader implications of TNF inhibitors on various immunotherapies and the potential for modulating the immune response.

Discussion on TNF and Immunotherapy

- Dr. Deichaite explains the lack of studies showing that blocking TNF affects the efficacy of checkpoint inhibitors, suggesting it may be a promising approach.
- The discussion touches on the complexity of modulating the immune response and the potential for TNF inhibitors to enhance immunotherapy efficacy.
- Brad Power asks about the intersection of Ida's research with other immunotherapy approaches, such as CAR-T and neoantigen vaccines.
- Dr. Deichaite and Brian McCloskey discussed the complementary nature of their research and the potential for combining TNF inhibitors with other immunotherapies.

Questions and Answers

- Dr. Deichaite answered questions about the stage of prostate cancer patients suitable for TNF treatment and the potential for combining TNF inhibitors with androgen ablation blockers.
- The discussion included the importance of diagnostics and tests to personalize treatment and the potential for collaboration with BostonGene for larger data sets.
- She emphasized the need for further research to validate the findings and explore the potential of TNF inhibitors in clinical trials.

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TRANSCRIPT

Brian McCloskey: We are very happy to have Dr. Ida Deichaite from UC San Diego join us. Ida is an assistant Adjunct Professor of radiation medicine and applied science at UC San Diego. She's a molecular biologist by training, and she's been doing significant research on tumor necrosis factor, TNF, and its relationship to immune response in prostate cancer. Beyond that, she is the director of industry relations at Moores Cancer Center. Ida and I go back to, I think 2017, when she boldly asked me to present some hair brain idea that I had about how medicine should be approached, and we've been fast friends ever since then. She and her teams have been very helpful in terms of looking at my genomic and multiomic data to identify potential treatment options. She’s an amazing friend, and we are looking forward to having her talk.

This is not medical advice, it is information you may take to your medical team. Always consult your physician.

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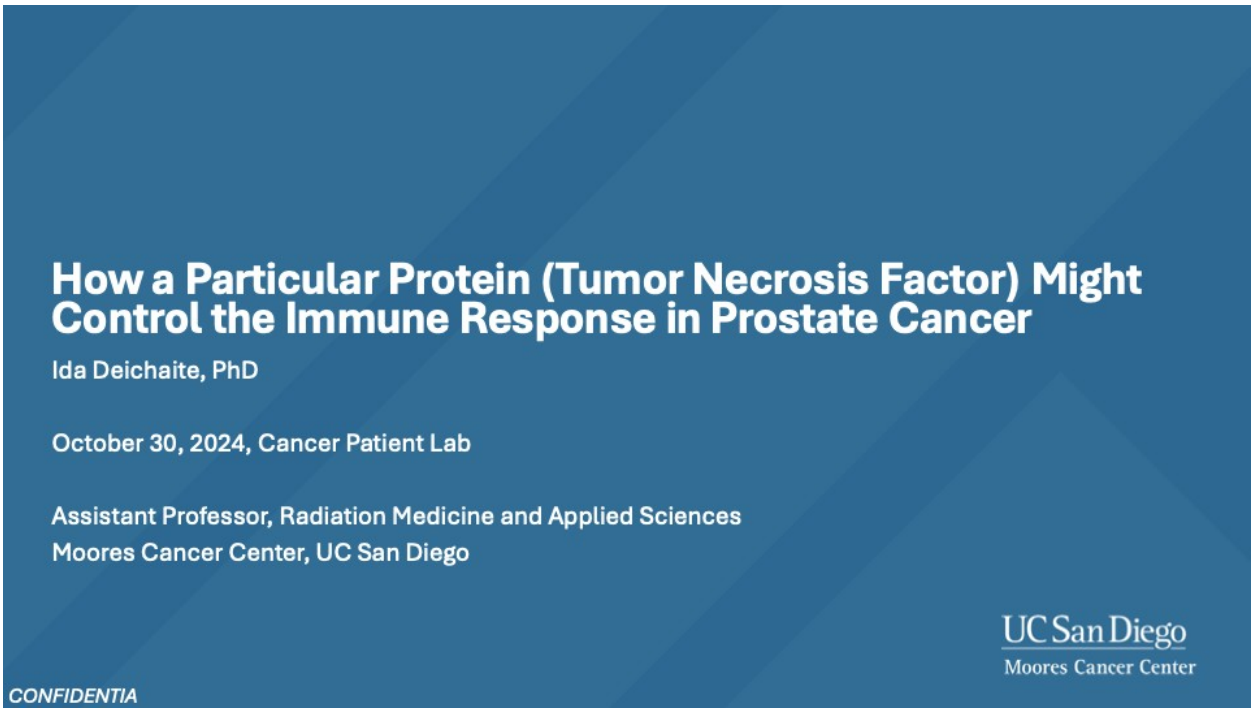
Ida Deichaite: I focus more on the patient group here, because I figured that if we want to get deeper into science, we can just connect with the researchers here. I just wanted to bring everybody on the same page and think about the research we're doing, or specifically by my team, and how we can think about it in terms of translational approaches (bringing research from the lab to patients), and how we can take it to patients.



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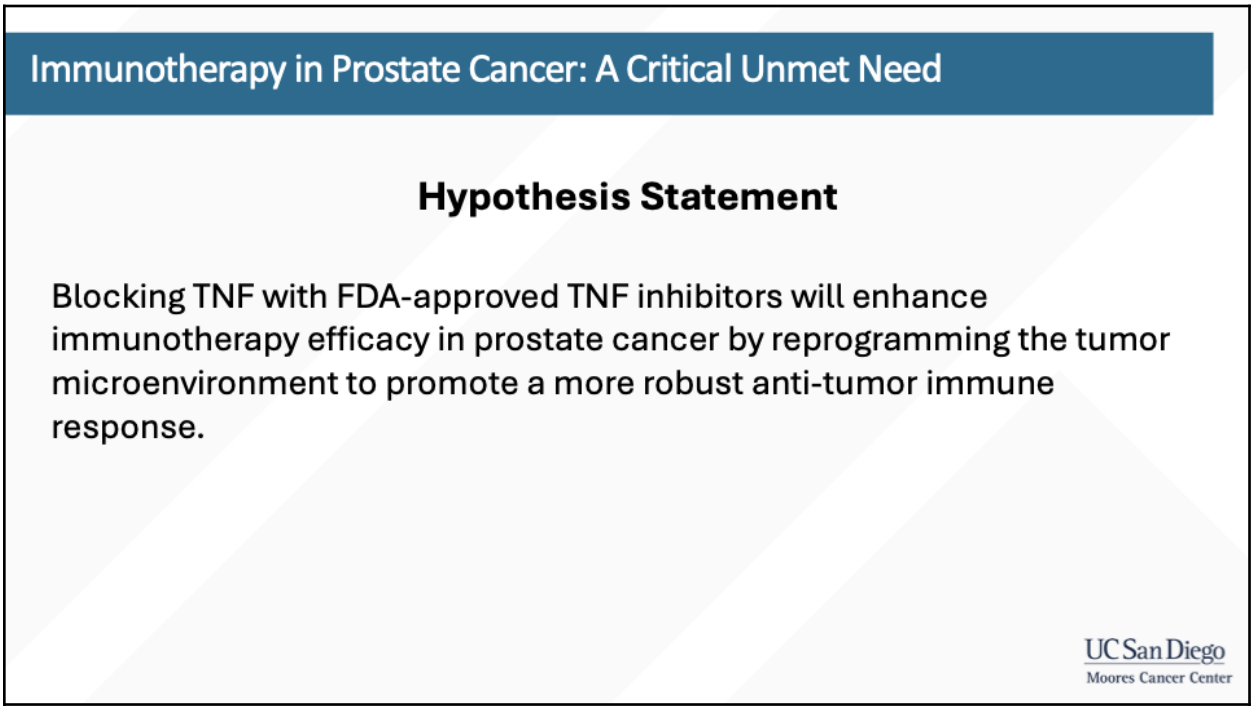
The title, which I discussed with Brad, focuses on TNF – how a particular protein, in this case, tumor necrosis factor, might control the immune response in prostate cancer. We'll just organize ourselves around this.



I will start with the hypothesis, because we'll repeat the hypothesis several times as we get to know a little bit more about the data. As you can see it's clearly stated, we think that blocking TNF with existing, approved FDA drugs may enhance immunotherapy efficacy in prostate cancer by reprogramming the tumor microenvironment to promote a more robust anti tumor immune response. And we really are thinking about checkpoint inhibitors, which are widely available and FDA approved. And this really comes from the clearly unmet need we are wanting to understand, how can we improve the efficacy of checkpoint inhibitors in prostate cancer? And we'd try to understand what makes prostate cancer cold. So the hypothesis is really to address this unmet need.

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Tumor Necrosis Factor (TNFα)* is a Key Cytokine that Plays a Crucial Role in Inflammation and Immune System Regulation

TNF is produced primarily by immune cells such as macrophages, T cells, and natural killer cells in response to infection, injury, or other immune stimuli

Initiating Inflammatory Response: TNF acts as a signaling molecule, initiating and amplifying the inflammatory response. It activates other immune cells to produce additional cytokines and chemokines, creating a cascade of signaling that recruits more immune cells to the site of inflammation.

Activating Endothelial Cells: TNF induces endothelial cells (cells lining blood vessels) to express adhesion molecules, which help immune cells attach to blood vessel walls and migrate into inflamed tissues. This increases the local concentration of immune cells in the affected area, allowing them to fight infection or repair tissue.

Increasing Vascular Permeability: TNF increases the permeability of blood vessels, allowing immune cells, fluid, and proteins to exit the bloodstream and enter the tissues. This causes the redness, heat, swelling, and pain typical of inflammation

Inducing Apoptosis in Infected or Cancerous Cells: TNF can signal cells to undergo apoptosis (programmed cell death), helping to eliminate cells that are infected, damaged, or cancerous.

Role in Chronic Inflammation and Autoimmune Diseases: While beneficial in the short term, chronic or excessive TNF production is associated with several autoimmune and inflammatory diseases, such as rheumatoid arthritis, inflammatory bowel disease, and psoriasis. In these conditions, TNF contributes to tissue damage and prolonged inflammation.

**TNFa will be listed as TNF in this presentation*

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So just to bring everybody on the same page by refreshing our understanding of TNFs function in inflammation. Inflammatory processes are complex, and TNF plays multiple roles within them. It acts on various pathways initiating and regulating the body's response. So let's explore these effects in more detail. TNF is implicated in the inflammatory response. TNF appears to be sort of at the head of the inflammatory cascade, and it initiates a lot of a signaling that eventually becomes kind of a chain effect and accomplishes what needs to be done to address pathogens or any sort of cell problems that need to be removed from the system. So that includes initiating the inflammatory response just like I said. What's interesting is it’s heavily involved in activating endothelial cells, helping the peripheral immune cells reach their targeted tissue size. So it helps to localize the immune cells to the actual tissue where pathogens took over, or there is any other reason to remove infection in those tissues. So it also increases vascular permeability, further enabling the reach of peripheral immune cells into the affected areas. It turns out that TNF may also be involved in clearing the inflammatory response, and that's another effect that TNF has. When TNF is over expressed for a period of time. It often leads to autoimmune diseases, and that's kind of a place where TNF has been targeted successfully.

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TNF Inhibitors are Widely Used in Clinic

- FDA approved TNF inhibitors are widely used to treat inflammatory and autoimmune conditions such as rheumatoid arthritis, Crohn's disease, and psoriasis. Some of the major drugs in this category are:
- **Humira (adalimumab)** - Approved in 2002, Humira has become one of the best-selling drugs globally, widely used across multiple autoimmune diseases, market share of $13 billion globally.
- **Remicade (infliximab)** - Approved in 1998, Remicade was one of the first TNF inhibitors and is widely used in treating rheumatoid arthritis and Crohn's disease.
- **Enbrel (etanercept)** - Approved in 1998 as well, Enbrel is frequently prescribed for rheumatoid arthritis, for psoriasis and ankylosing spondylitis inflammatory conditions.
- Biosimilars like **Amjevita** and **Cyltezo** entered the market, offering cost-effective alternatives to patients and healthcare systems.

TNF inhibitors have a well-established safety profile. While generally safe, the primary risks involve common infections, which can be managed effectively with regular monitoring

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We know today that there are many TNF inhibitors which are widely used in clinics in autoimmune diseases. So HUMIRA is one of the key biologics that are used widely, but the first approvals came in as early as 1998 and obviously these are widely used drugs. They capture a huge market share, 15 billion globally. Today, we already have biosimilars coming along, such as Amjevita and Cyltezo, so access to these drugs is pretty well addressed. And what we know from using those drugs or those drugs being in the clinic from 1998 is that they have a well established safety profile, and they're considered to be generally safe. While there are some risks, they can be managed with regular monitoring.

The Dual Nature of TNF: Inflammatory and Immunosuppressive Function

- TNF is primarily known for its pro-inflammatory role.
- TNF can have immunosuppressive functions that contribute to immune regulation and homeostasis, preventing overactive immune responses that could damage healthy tissue

Prostate Cancer Exploits TNF-Driven Immunosuppressive Mechanisms, Leading to a “Cold” Tumor Microenvironment and Reduced Response to Checkpoint Inhibitors

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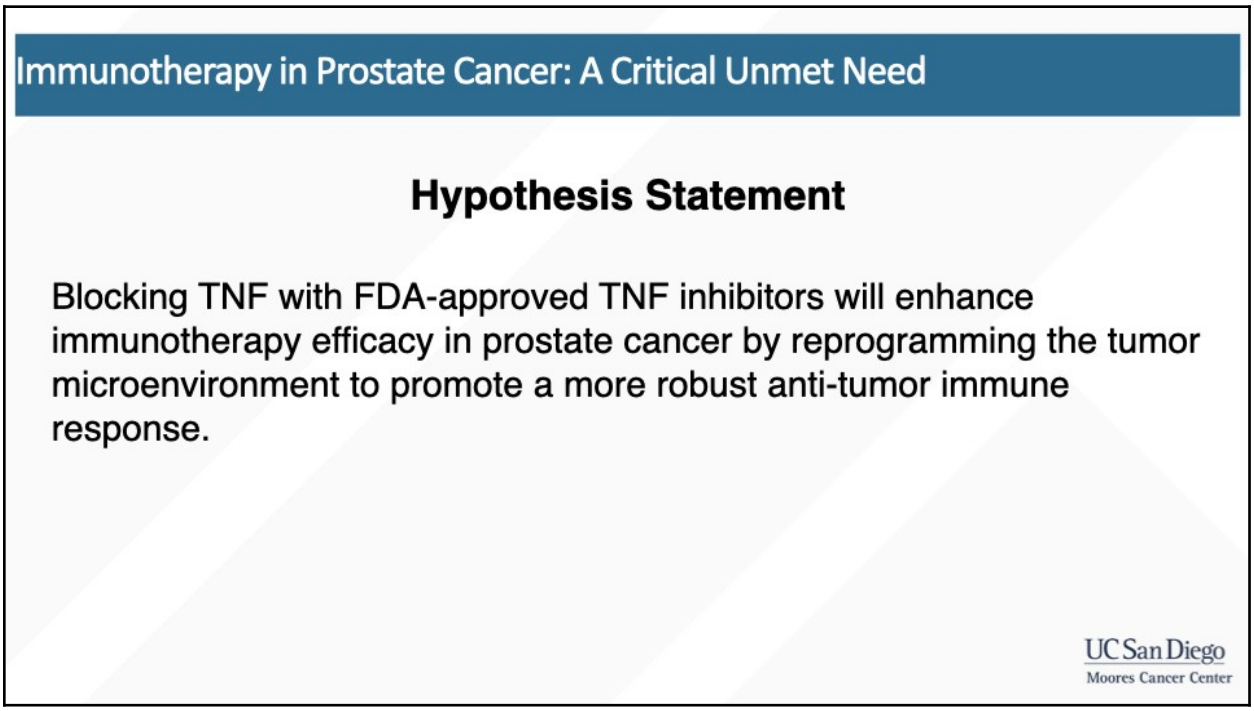
So what's interesting to me, and where I'm focusing, is that it turns out that at some point the inflammatory response needs to be resolved to allow tissue healing. Currently, it's a very hot field of research, and the mechanisms driving this transition from inflammation to healing are being studied. However, what's really interesting to me is that TNF may play a role in this process too, and so TNF has a dual function, it has an inflammatory function that's very well understood, but it may also have an immunosuppressive function that is less understood. Lately I see more and more literature on that, and more and more research on that. So what we are thinking, based on our genomic study, is that in prostate cancer, the prostate exploits TNF or hijacks TNF driven immunosuppressive mechanisms, and that contributes to cold tumor microenvironment and reduced response to checkpoint inhibitors.



And from there, we are again restating our hypothesis that perhaps blocking TNF, and in this case, blocking the immunosuppressive function of TNF, with FDA approved TNF inhibitors will enhance checkpoint efficacy in prostate cancer, again, by reprogramming the tumor microenvironment to promote a robust anti tumor response.

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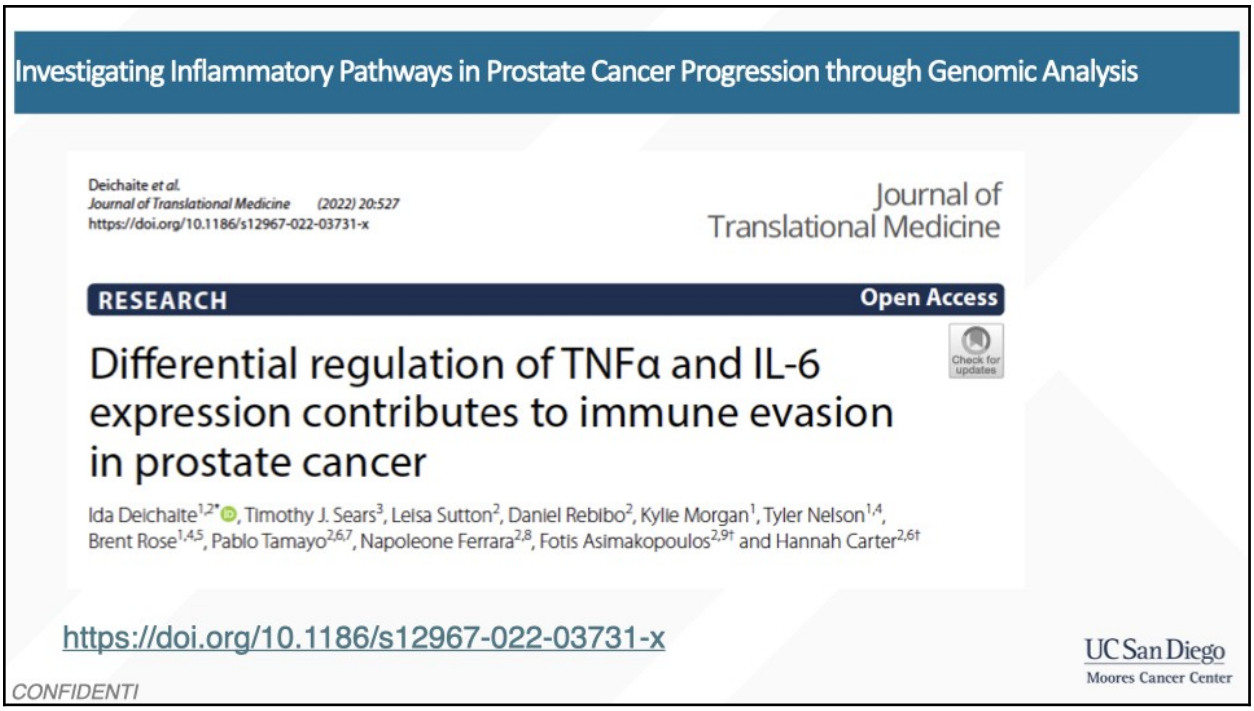
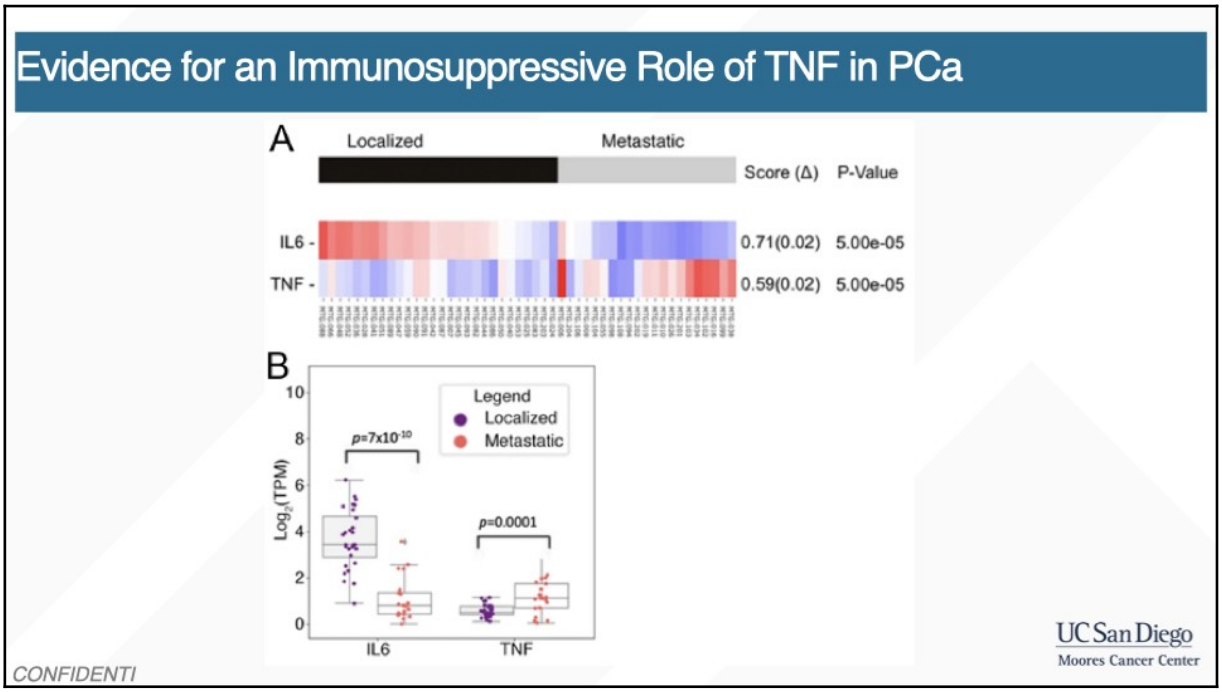


So how did we come up with all this? That actually stems from a bioinformatic study of prostate cancer genomics, and we focused on RNA seq genomics. When we started this study, we were completely agnostic. We wanted to know what happens with the inflammatory microenvironment in prostate cancer, because there are many publications suggesting that inflammation plays a role, but there isn't really a consensus of what happens, how inflammation contributes to the prostate cancer. And so we dove into this area.



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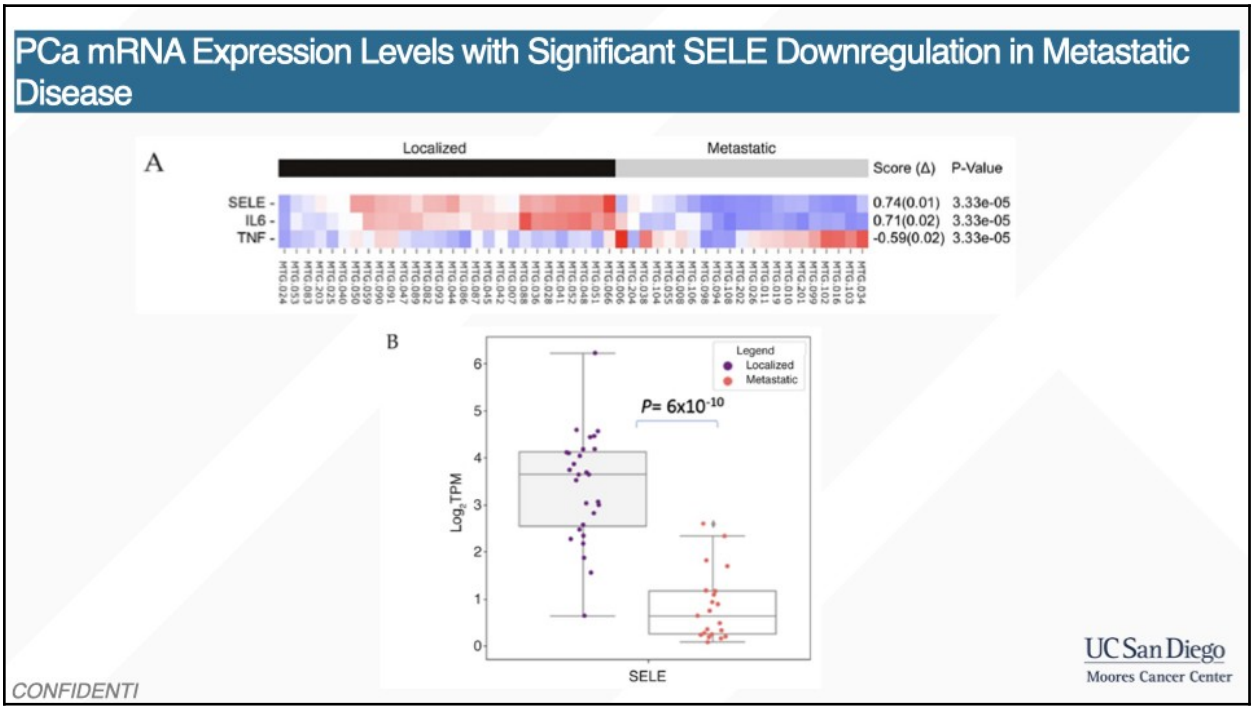
We looked at the RNA seq data from prostate cancer patients from the cohort that we have at the Moores Cancer Center. Our data comes from Tempus, while Tempus provides the reports to the physicians, they also upload the raw data from the sequencing of the pathology samples. So all our data is really patient data. I trust patient data more than experimental models in prostate cancer, especially because they happen to be rather weak. So this data is directly from prostate cancer patients. So what we did is to really look at the inflammatory landscape, we binarized our question, and basically what we said is, what is the difference between localized and metastatic disease? So we could see the progression of disease and how the inflammatory proteins are contributing and how they change with the disease progression. And what was really interesting to me was that then we had a look at TNF, and in this case, we had 49 patient samples. Then we looked at TNF, we saw that TNF is consistently expressed in prostate cancer with some increase, there's an increase with progression. So that was kind of interesting, that TNF is there, and what does it do? But what became even more interesting to me is, as disease progresses, the IL6, which is a cytokine that is directly activated by TNF is regressing. And that was kind of unusual to me. I used to be more involved in immunology studies. And my kind of, you know, sort of the dogma I had in my mind, is that when TNF is high, IL6 is high. So when I saw this reverse relationship I was wondering if it's an artifact. We looked at patient charts to see if there is something specific about this, maybe steroids, etc, but we couldn't detect the connection between treatment and the IL6 expression. And so we decided to focus specifically on the bioinformatics data and pursue it. in the A panel, you see the heat map, and you can see that IL6 is highly expressed in localized disease, but seems to regress in metastatic disease, while in TNF, it kind of increases with disease progression, just like I said. We also did box plots, and you can see that there is a high statistical significance and the differentiation within IL6 levels and TNF levels with disease progression in prostate cancer.



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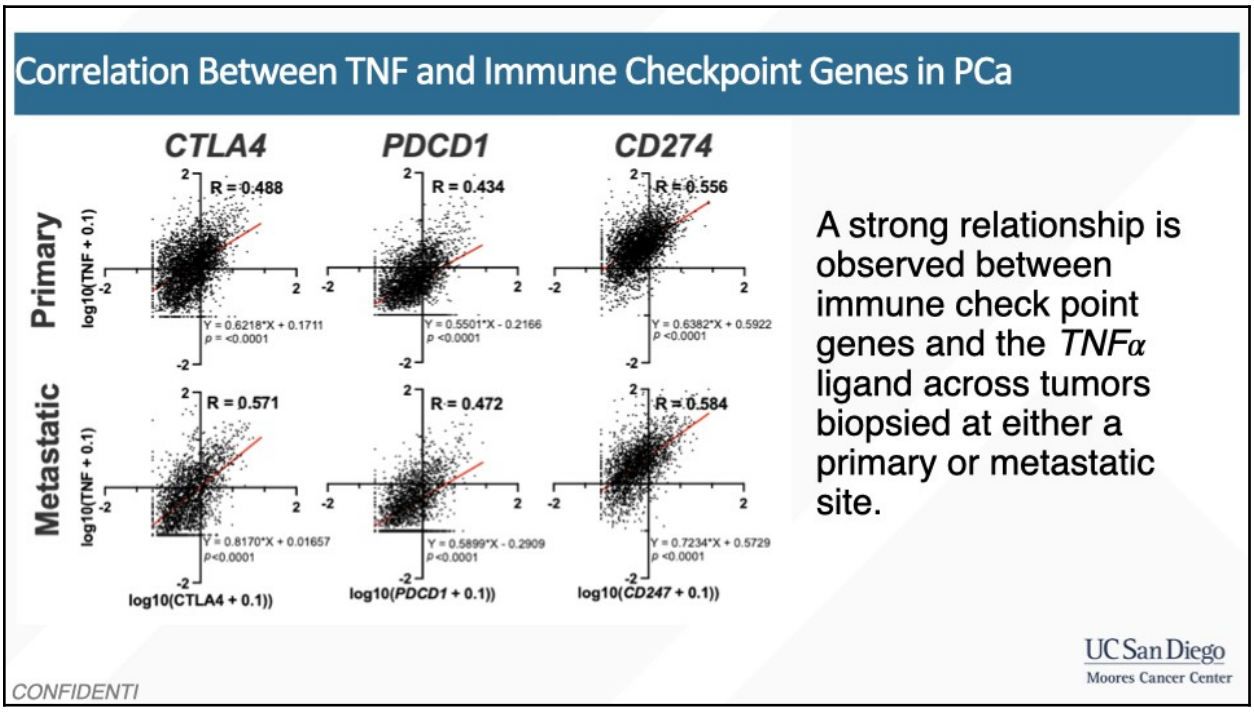
So then we had a look at other cytokines that are activated by TNF, one of them that stood out was E-selectin. So again, you see a similar sort of distribution of E-selectin with disease progression. It just seems to disappear in metastatic disease. However, it's expressed in the localized disease. It's very similar to IL6, and here we have sort of underneath that, we have the TNF expression and the box plot also just of E-selectin shows a clear statistical significance in differentiating the levels of expression of E-selectin with disease progression.



So then we had a look also at TNF correlation with checkpoint inhibitors, and we did the Pierce linear regression analysis. And what's interesting is we saw that, indeed, there is a strong relationship between TNF and expression of checkpoint inhibitors. This study was done by Caris, so they used a large cohort of patients, 1000s of patients, and there is definitely a statistically significant effect between TNF expression and checkpoint expression. So what it tells me is probably that the cold environment that we see in prostate cancer is not due to the lack of checkpoint expressions. So something else is going on. The checkpoints are there, but somehow they're not being engaged by checkpoint inhibitors to deliver efficacy.

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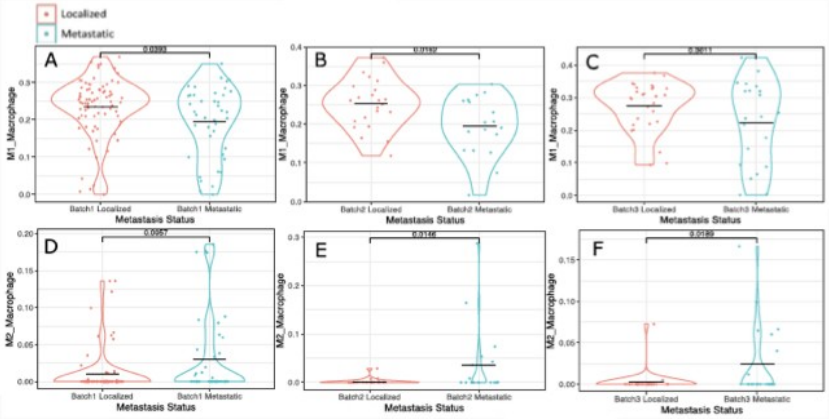
M1 and M2 Macrophage Infiltration of Localized vs Metastatic Patients in Our Study

A custom signature matrix was generated using single cell RNA-seq gene expression profiles in CIBERSORTx. This custom signature matrix of pre-metastatic and metastatic prostate immune cell populations was then applied to each of the RNA-seq cohorts to estimate immune cell infiltration via the “Impute Cell Fractions” module from CIBERSORTx.



Deichaite I, Sears TJ, Sutton L, Rebibo D, Morgan K, Nelson T, Rose B, Tamayo P, Ferrara N, Asimakopoulos F, Carter H. Differential regulation of TNFα and IL-6 expression contributes to immune evasion in prostate cancer. J Transl Med. 2022 Nov 12;20(1):527. doi: 10.1186/s12967-022-03731-x. PMID: 36371231; PMCID: PMC9652804.

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We used CIBERSORTx to look at the immunosuppressive immune cells in our patient cohort.. And that was actually published in our paper two years ago. You can see that we were able to detect the disappearance of M1 macrophages that drive inflammation and a higher level of M2 macrophages, just by doing a CIBERSORTx.

Low SELE Expression Correlates with a ‘Colder’ Tumor Micro Enviroment (TME)

SELE-High(Q4) vs SELE-Low(Q1) tumor microenvironment (TME) cell fractions using quanTIseq, a published method for RNA-based deconvolution of cell fractions – the median cell fraction for each immune cells subgroup is below.

Feature	Median Cell Fraction, Primary, SELE Q4	Median Cell Fraction, Primary, SELE Q1	Q4/Q1 fold change	Feature	Median Cell Fraction , Metastatic, SELE Q1	Median Cell Fraction , Metastatic, SELE Q4	Q4/Q1 fold change
quanTIseq-Macrophage M2	7.86%	6.96%	1.13	quanTIseq-Macrophage M2	6.99%	6.11%	1.14
quanTIseq-Neutrophil	6.66%	4.98%	1.34	quanTIseq-Neutrophil	8.19%	5.13%	1.60
quanTIseq-B cell	4.87%	3.96%	1.23	quanTIseq-B cell	4.86%	3.34%	1.46
quanTIseq-NK cell	4.53%	3.90%	1.16	quanTIseq-NK cell	3.79%	3.25%	1.17
quanTIseq-Myeloid dendritic cell	2.00%	1.17%	1.71	quanTIseq-Myeloid dendritic cell	2.09%	1.29%	1.62
quanTIseq-T cell regulatory (Tregs)	1.45%	0.49%	2.96	quanTIseq-T cell regulatory (Tregs)	1.61%	0.29%	5.55

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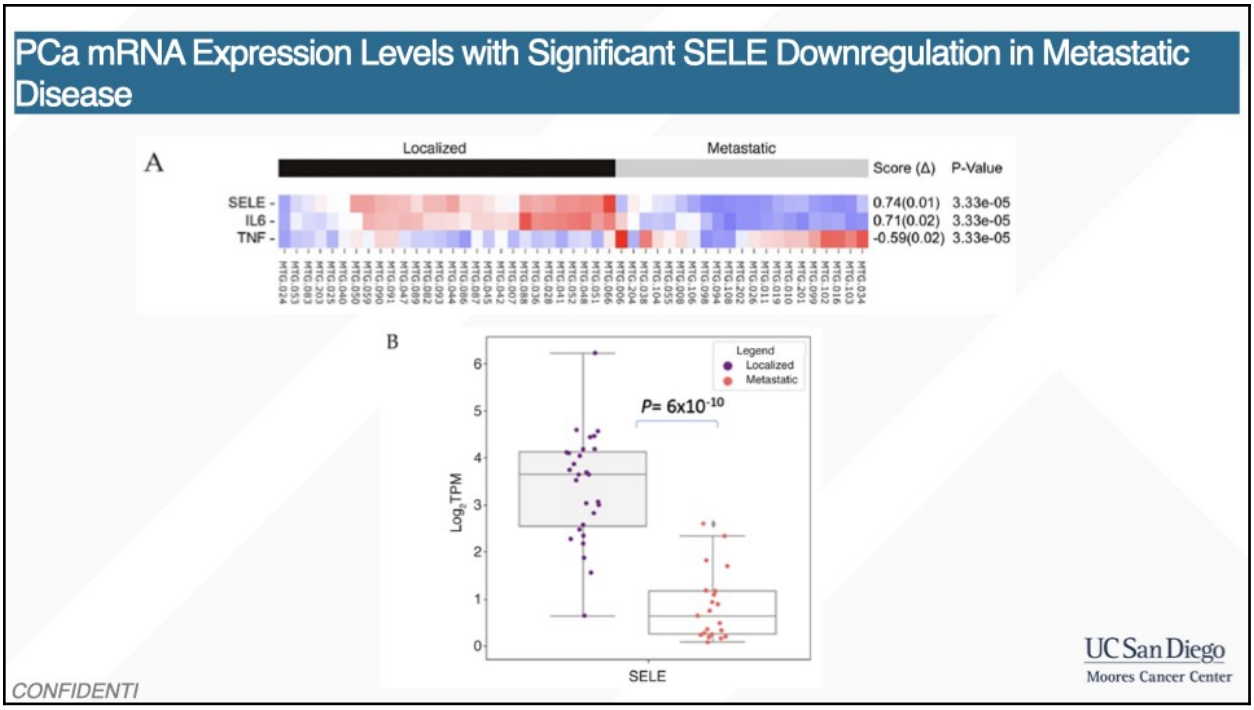
So recently, I connected with Caris and worked with Andrew. And this is the table that Andrew generated based on Caris’ data. Now, here we're including about 5000 patients, as opposed to 49 patients that we analyzed in our paper, and it looks that Andrew’s results are rather consistent with ours, showing that prostate cancer indeed shows in these samples that there is a reduction of the immune cell subgroup that should be helpful with checkpoint inhibitors.



I want to go back to E-selectin, because I'm really interested in E-selectin and what role it plays. Just to remind you again that E-selectin seems to be disappearing, or the level of the expression of E-selectin, based on RNA Seq analysis, seems to be highly reduced with disease progression, and so we kind of dove in to try to find association between E-selectin and its functional activity in prostate cancer.

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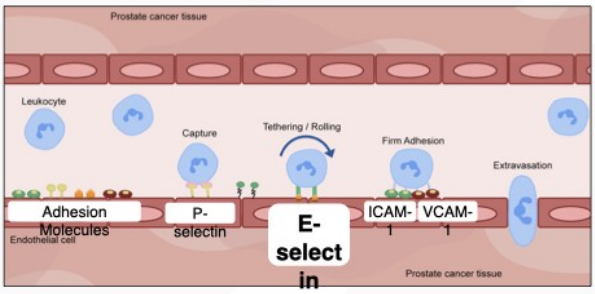
Impact of E-Selectin Downregulation on Immune 'Cold' Prostate Cancer: Limited Infiltration of Key Immune Cells for I-O Response



E-selectin downregulation may limit immune cell infiltration, contributing to immune evasion and poor anti-tumor immune response.

E-selectin localized activation is driven by inflammatory TNF and IL-1b signaling in response to infection/ cell damage to enable peripheral immune cells infiltration of the affected site.

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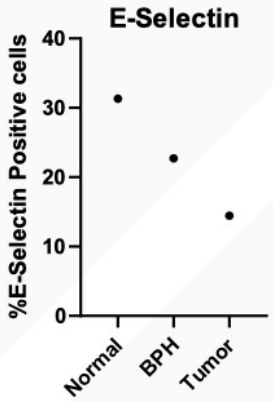
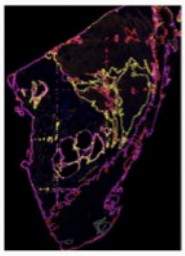
E-selectin is an interesting molecule, and kind of a really good molecule to study, because E-selectin is very specific for endothelial cells, it's being only expressed on the vasculature. We're looking specifically at the vasculature within the prostate cancer tumor, and E-selectin is activated mainly by TNF as well as IL-1a/b so it kind of gives us a really good picture of the play of TNF in this case.

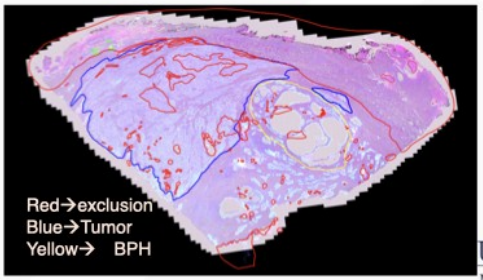
Bioinformatic Analysis and Histology Highlights the Differential Expression of E-Selectin in Different Areas of the Tumor

Data from Visiopharm excel file





BPH Dapi	338603
bph e-selectin pos	99493
BPH e-selectin pos (%)	0.227
BPH total nuclei	438096
normal dapi	78512
normal e-selectin pos	35788
Normal e-selectin pos (%)	0.313
Normal total nuclei	114300
Total dapi	774455
total e-selectin pos	192545
Total nuclei	1.95e+06
tumor dapi	367008
tumor e-selectin pos	61925
Tumor e-selectin pos (%)	0.144
Tumor total nuclei	428933



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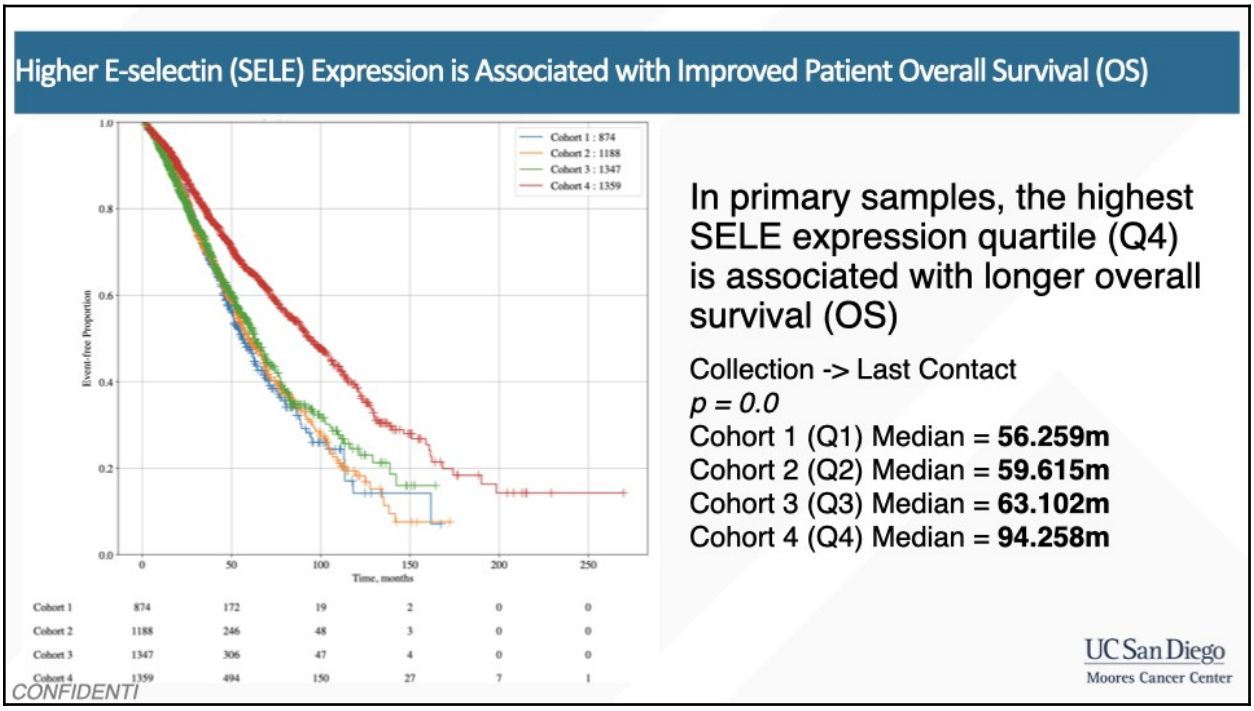
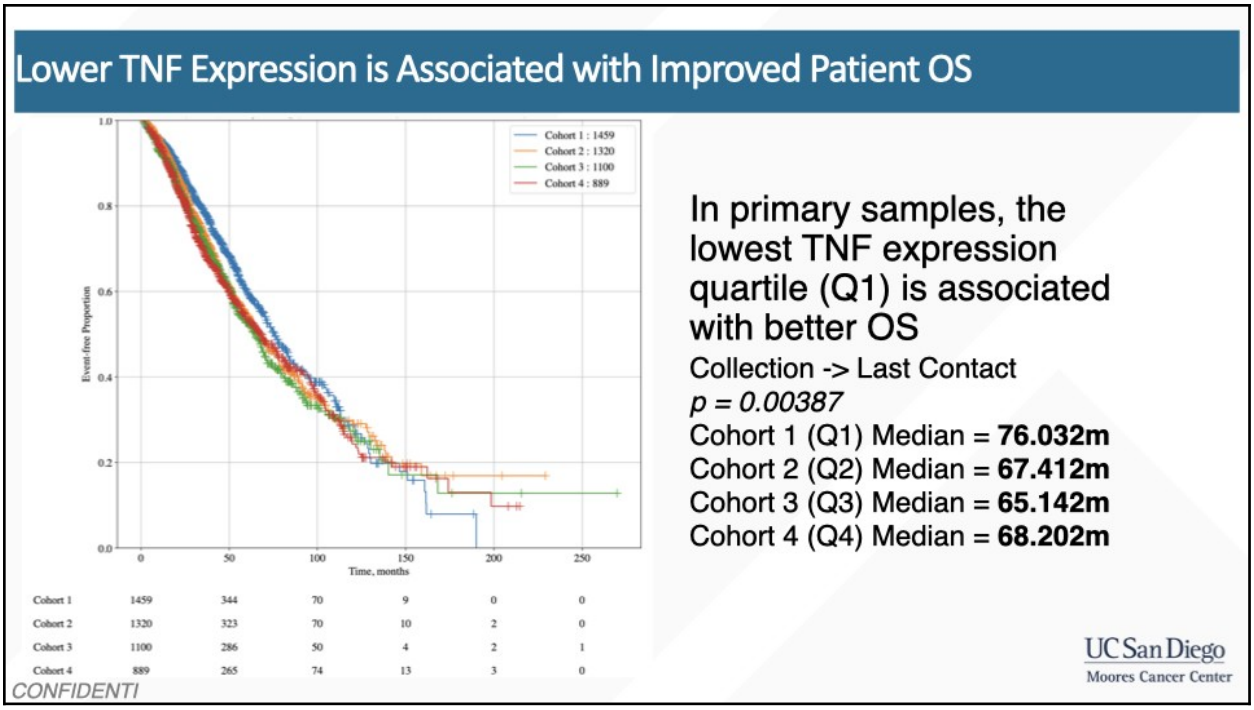
“How a Particular Protein (Tumor Necrosis Factor) Might Control the Immune Response in Prostate Cancer” (Ida Deichaite, PhD) [#118]

So the next slide is interesting, it took me only one year to get this data. So apologies for taking so long, but I was trying to understand how to explore the E-selectin expression in the tissue, looking at proteins, because we only had RNA Seq data, does RNA Seq translate into protein data? Which is what we really want to understand. And I was trying to understand what is the best experimental system to research this effect? I didn't want to go into my mouse experimental systems, because our data is from humans, and we wanted to reproduce it in humans. So I came up with the following idea. I learned that prostate cancer in some patients, shows up as a couple of different comorbidities, among them that the patient may have a tissue spot that is BPH, so it's kind of more an inflammatory environment, benign hyperplasia, as well as a tumor right next to it in the same patient. And so we went to the pathologist and asked to identify patients who have this type of pathology findings, and then we send it to Wash U to stain for E-selectin protein expression. We kind of reduce the heterogeneity of the data, this is the same patient governed by one set of genomics. And indeed here you see on the left that we see a significant reduction in E-selectin protein in the tumor tissues. So the normal tissue has the most E-selectin protein, BPH has a slight reduction. And then in the tumor, you see the reverse: significant reduction in the protein E-selectin which we know is usually associated with endothelial cells. And in this case, we think it's associated with a microvasculature within the tumor tissue.



So then, we went back to the Caris data to try to understand if E-selectin expression levels may correlate with overall survival response in patients. Andrew, who is working with us, produced this data, and what it shows is that a high expression of selectin is associated with significantly improved outcomes in overall survival. So that data is encouraging. Of course, you know, it’s just correlative data, but it looks like the highest quartile, the patients who express the

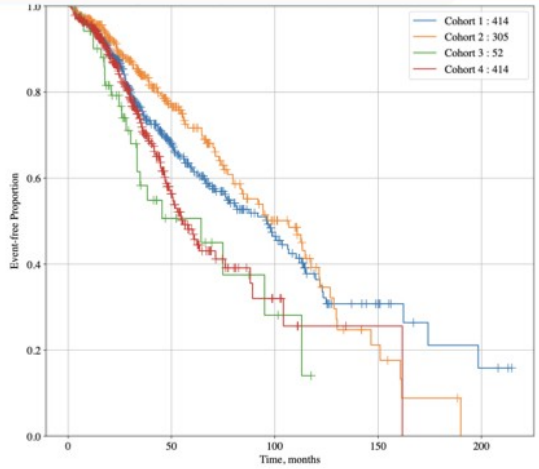
highest level, which is quartile 4 in this case, the red plot, shows an improvement in overall survival.



So then we looked at TNF and it was somewhat consistent and encouraging, that the less TNF the better. So while it's not such a clear isolation as with E-selectin expression, it still is consistent, or at least it's consistent with our hypothesis that TNF in prostate cancer is playing an immunosuppressive role, and therefore the less of it we see, the better it is in terms of overall survival. So Q1 is the quartile of patients that have the lowest expression of TNF.

“How a Particular Protein (Tumor Necrosis Factor) Might Control the Immune Response in Prostate Cancer” (Ida Deichaite, PhD) [#118]

OS from Biopsy Across SELE/TNF Q1/Q4 Combinations



Cohort 1	414	156	49	11	3
Cohort 2	305	110	36	6	0
Cohort 3	52	11	3	0	0
Cohort 4	414	68	7	1	0

Among primary samples, high SELE (Q4)/low TNF (Q1) was associated with the longest OS

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p = 9e-05
Cohort 1 (SELE Q4/TNF Q4) Median = **96.331m**
Cohort 2 (SELE Q4/TNF Q1) Median = **106.76m**
Cohort 3 (SELE Q1/TNF Q4) Median = **64.352m**
Cohort 4 (SELE Q1/TNF Q1) Median = **55.601m**

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So then Andrew looked at the dual expression of E-selectin in TNF, he found that high E-selectin expression in patients and low TNF show the best overall survival outcomes.

TNF Inhibitors in Combination with Checkpoint Inhibitors in Clinic

TNF inhibitors may enhance checkpoint immunotherapy effectiveness

- In advanced melanoma clinical trial TICIMEL, TNF inhibitors combined with checkpoint inhibitors increased tumor regression rates and improved immune activation in tumors, especially by enhancing CD8+ T-cell activity.
- TNF blockers, like infliximab reduce immunotherapy adverse effects, including colitis, without compromising the immunotherapy's anticancer efficacy.
- Published reports show that TNF inhibitors might prevent secondary checkpoint expression on immune cells, improving treatment durability and reducing immunotherapy toxicity.

Dufau C, Genais M, Mucher E, et al. Ceramide metabolism alterations contribute to Tumor Necrosis Factor-induced melanoma dedifferentiation and predict resistance to immune checkpoint inhibitors in advanced melanoma patients. *Front Immunol.* 2024;15:1421432. Published 2024 Jul 29. doi:10.3389/fimmu.2024.1421432

Montfort A, Dufau C, Colacios C, et al. Anti-TNF, a magic bullet in cancer immunotherapy?. *J Immunother Cancer.* 2019;7(1):303. Published 2019 Nov 14. doi:10.1186/s40425-019-0802-y

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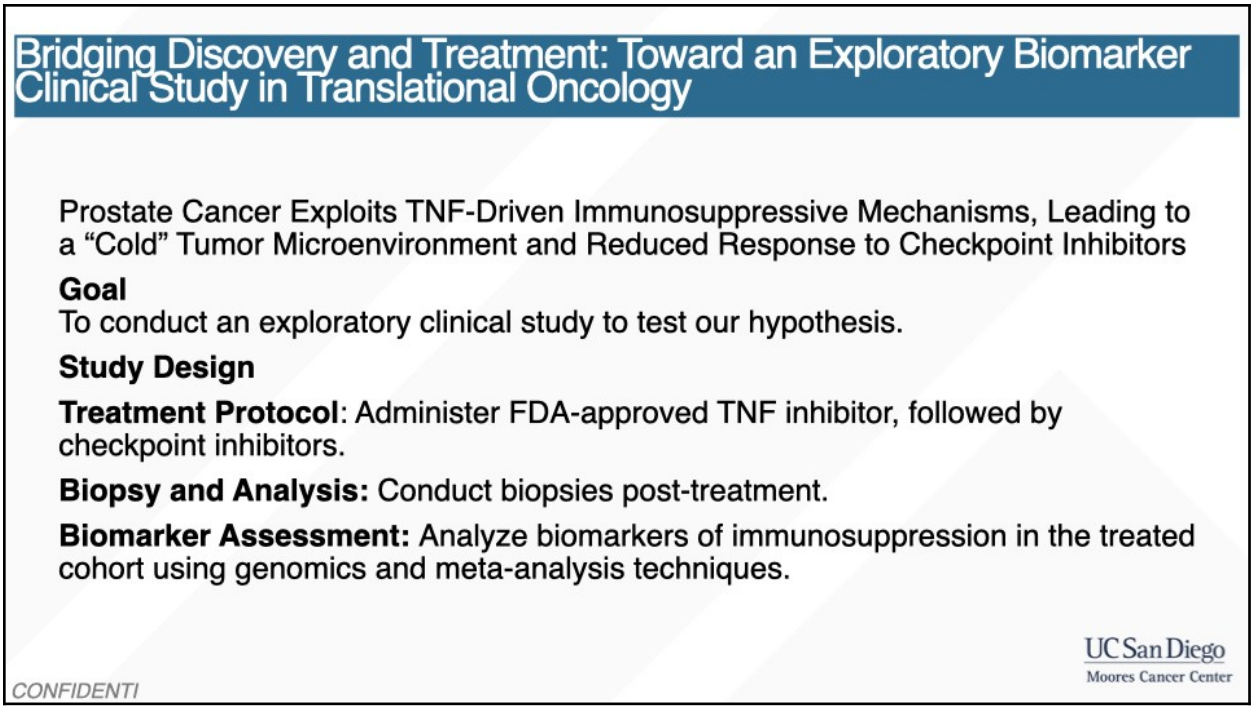
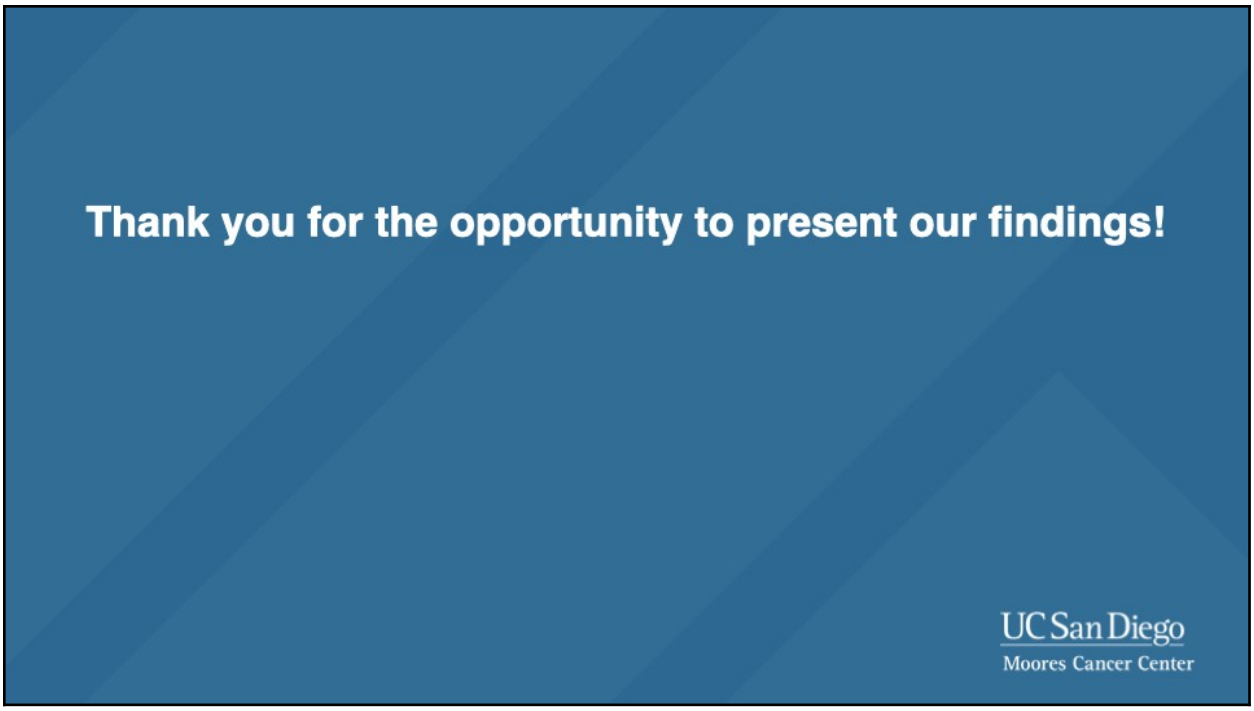
We looked into literature, we wanted to see if TNF inhibitors were used in combination with checkpoint inhibitors, just to see what has been done in the clinic with this particular

combination. And interestingly enough, there is a clinical trial study TICIMEL, the study design included TNF. And the reason for including TNF with checkpoint inhibitors was to reduce the adverse effects from checkpoint inhibitors. When the data was analyzed, the group that had TNF inhibitors in combination with checkpoints showed better efficacy, although it's a very small cohort of patients, but that is encouraging. Also, it was found in other studies, the TNF blockers reduce the immunotherapy adverse effects, but they do not compromise the efficacy of checkpoints. And then there is emerging data suggesting the TNF inhibitors may prevent secondary checkpoint expression.



The key point of our presentation and work is to propose an exploratory clinical study to test our hypothesis. We are considering an approach that combines TNF inhibitors, which have a good safety profile, with checkpoint inhibitors. This treatment would be administered prior to biopsies, followed by post-treatment analysis of biomarkers of immunosuppression and specific immune cells associated with immunosuppression.

“How a Particular Protein (Tumor Necrosis Factor) Might Control the Immune Response in Prostate Cancer” (Ida Deichaite, PhD) [#118]



So that's the presentation I had for you today, and I'd really appreciate questions and ideas you may have.

Question and Answer Section

Brian McCloskey: Robb Owen: You teed up some questions in the chat. One is regarding zinc, and then the other is regarding Cimetidine. Do you want to elaborate on those?

Robb Owen: I was curious if, in doing your research, whether you look at the effect that zinc has on epithelial cells and how it protects them from apoptosis due to TNF? Has that ever been a consideration?

Ida Deichaite: In my case, I would need to find the patients who receive zinc and look at the genomics. It would be a really interesting study, but I have not done that.

Robb Owen: I'm finding effectiveness with it. I assist cancer patients worldwide with some of these methods I'm sharing with you right now. How are we able to control TNF with Cimetidine? You control its production, throw in a couple of other…basically, we're regulating neurotransmitters along with enhancing, doing portions to enhance the immune system through diet and supplements, with great success. And just wanted to see if, if any of this had crossed your mind. I know you're very focused on the components you're working on, but these may be something interesting for you to consider.

Ida Deichaite: Thank you. I think it's very interesting, because when we did our genomics, one of the striking findings was we saw a real significant elevation of GABA, and we really can, you know, drive it forward, because it's kind of complicated, but we were very curious about that. Clearly, there is something going on with it.

Robb Owen: Oh, absolutely. And it influences PI3K, AKT, which regulates VEGF and transforming growth factor, beta 1 (TGF-β1). You put the combination of these together with Cimetidine and it enhances the effects of the Prochlorperazine and the Alprazolam to really very effectively manage those and several other signaling pathways and growth factors. So, I think it'd be something that, if you have time or the desire, it’d be worth looking into.

Brian McCloskey: Robb, maybe just explain to Ida a little bit about your background. I think we already set you up with Saul Priceman from Salisbury.

Robb Owen: I’ve got a meeting with him here in a couple of weeks, so I appreciate that.

A little over a year ago, I was diagnosed with stage IV head and neck squamous cell carcinoma, I figured out in the middle of my treatment a better way to treat cancer and my tumor resolved in two weeks. I followed most of this protocol, except I was not taking the Cimetidine in my case, the Prochlorperazine and the Alprazolam, had a very limited diet supplement regimen. When my doctors couldn't explain why I healed so quickly, I decided to reverse engineer it and figure it out myself, and by doing so, I was able to stumble basically across these things, and I've been building upon that knowledge across all types of cancers. We're having significant success with

glioblastomas, with stage four, non small cell lung cancers. Just had a stage three cervical cancer with meds to lymph nodes have 70% regression of the tumor in three and a half weeks with standard of care. I ended my treatment after three weeks, it was supposed to be a seven week treatment. I'm still NED, pretty healthy besides the cold, and so I've been working with integrative oncologists worldwide. I've got, I'm working with University of Utah, School of Medicine, Chris, Greg and I are working through this together, going to meet with Saul, that Brian was kind enough to introduce me with USC shortly. So I work with an integrative oncology group in India where we're treating patients with these methods. I've talked to Dr Michael Castro. He's in agreement on some of the research that I've found, and basically I'm looking at it from micronuclear deficiencies, driving all DNA mutations, and then by managing it through diet and supplements on that side, you build the immune system up to help fight your cancer cells, and by managing the neurotransmitter side to signaling pathways with commonly prescribed medicines that are mostly given to cancer patients, the anti medics groups. You get the correct combination of those based on gender, pre existing conditions, the specific type of signaling pathways that the cancer uses to grow. When you put that combination together, we're having remarkable success in managing cancer better.

Brian McCloskey: Thanks for the background, Robb. Those are really amazing stories. Ida, we can certainly take that offline.

By the way, you mentioned Dr. Michael Castro, who is at Cellworks. I reached out to him because I know that he has an interest in this. Unfortunately, he, I believe, is in clinic today, so he couldn’t make it, but Ida will probably share this afterwards, and he might be somebody else that you might be interested in speaking to. He's a very talented physician.

Brad Power: Ida, you're focusing on the immune system dynamics. It's almost like an ecosystem, and there are accelerators and brakes, and you're trying to tune the immune system to work effectively in ways that would help the patient. Checkpoint inhibitors are the most well known immunotherapy, the one with the most experience. But there are also personalized neoantigen vaccines. There is also CAR-T and CAR-NK. There are other things on the horizon. There are other immunotherapies that would benefit from this.

I would share a personal experience: I had CAR-T two weeks ago, and I went through the whole cytokine release syndrome. I had to be hospitalized. It's normally done in the hospital. I was outpatient, but had to be hospitalized when I had the response that I did. I'm curious: There's the cold vs. hot tumor thing, mostly focused around checkpoint inhibitors, but you could also be modulating the immune system somehow. Sometimes you want to turn it up, sometimes you want to turn it down. So in my case, when I started having fevers, they would then do things to turn down the fever. But in general, they didn't want to do anything that would modify the inflammatory response. They want the CAR-T to run, and they want the near fever response, but not the over fever response. So it seems like being able to modulate the intensity of the immune response, or to channel the immune response, to tune the immune response, would be generally a great thing for any immunotherapy, not just checkpoint inhibitors.

Ida Deichaite: It's interesting, because I'm not aware of studies where they used TNF the way we have done, like a rigorous study where they showed that. The anticipation is that you want the inflammatory response, that the inflammatory response is helping to get rid of the tumor. So these are connected. But the surprise was when the rigorous study was done, it was shown that TNF, or blocking TNF, was not really in any way interfering with the response…so you can decouple the blocking of TNF and the efficacy of checkpoint inhibitors without really affecting the immunotherapy efficacy. So in this case, I'm aware of a study that has some scientific rigor, as opposed to just wanting the inflammation to be all out, which creates a lot of adverse effects. I think it's to be studied with CAR-Ts. It's probably a little bit more difficult to do than with checkpoints, because CAR-Ts are more individualized. Anytime you can use a drug versus cellular therapy it is much easier. I'm just aware of this data, which I find very interesting and not intuitive.

Brad Power: That took care of that question.

From the conversation with Saul Priceman last week, what would you see at the intersection of Ida’s research, and his research, because he was looking at bispecifics and CAR-T and the progress of immunotherapies generally in prostate cancer. How might you characterize the intersection of their research?

Brian McCloskey: At a really high level, it's just complementary. What Ida would do with the TNF strategy would simply complement what he's doing from a CAR-T perspective. I get these confused. There's an autologous CAR-T and then there's the, what's the other one? Ida help me? Somebody help me? Allogeneic? Yeah, allogeneic.. I know that he's pursuing both of those. He's also looking at multiple targets. So we talked about, rather than just going after, for example, a PSMA-targeted CAR-T, going after several different targets, just given the heterogeneity of the cancer. That's where his research is going. What Ida’s doing seems to be completely complementary to what he would be doing.

Brad Power: Does this change the paradigm at all? I know Rick Stanton, our dear colleague and friend, educated us about hot and cold tumors. Does this understanding of TNF at all modify that simple hot vs. cold distinction?

Ida Deichaite: That's a really excellent question, Brad. We're now looking at, what are the immune cells of that particular experimental system we designed where we can look in the same patient and look at differential expression, or differential presence of immune cells that are associated with immunosuppression. So cold versus hot tumor. We're now able to see what are the cells that show up in tumor versus BPH that's considered to be hot or that has these inflammatory factors associated with good responses to checkpoints and the stroma. That data is trickling in, and it's really interesting data. I'm just really trilled by it, but it's just too early to speak about it, so we'll write it up soon. But the real issue here is, can we connect TNF to all this, because obviously there are many different factors. We like TNF because it can be easily targeted, and it has a good safety profile. So we have hope that TNF will come through on this. But we keep researching this.

I saw a real interesting question. Somebody asked if we could add selectin to the mix. I don’t know who asked that, but I saw that question popping up.

Brian McCloskey: Yeah. Rick Davis did. Can you supplement increase selectin in a clinical trial, by adding selectin to the TNF inhibitor be a second variable to the test? What led you to selectin?

Ida Deichaite: Yeah, so that's an incredibly interesting question. We showed in our data that IL6 seems to be disappearing from the tumor tissue with disease progression, and so does E-selectin. However, IL6, in the past, was considered to be a target in prostate cancer. It was even taken to the point where it was considered to be a driver in prostate cancer, and it led up to a phase three study. So what does our data tell us? Is it an artifact or, you know, what are we dealing with? We discovered, and we actually found some validation through other people's research, that IL6 is highly prevalent in the periphery, so the peripheral blood has loads of IL6, but it doesn't trickle into the tissue, so the tissue seems to… the tumor tissue is stingy on IL6 expression, but IL6 is just all over the place in the blood, and the same is true for E-selectin. So the trick here is that E-selectin needs to be enriched within the tissue or the tumor, because that's where it really plays a role. The tumor also acts like, hey, you know, there is a problem, but it's not here, so go away. And E-selectin really drives the immune cells to get into that localized tissue where there is a problem. So now we don't have E-selectin, so T cells, B cells, etc, the important cells that engage the checkpoint inhibitors. The checkpoint inhibitors are there, they're highly expressed, so that's not an issue. They should work, right? We should be able to knock them out. But if we don't have the immune cells that knock them out, we really didn't do anything. So if we lose the T cells and the B cells that actually present the antigens or present the tumor antigens, and we reduce the checkpoint influence that creates an immune suppressive environment, we really haven't accomplished the goal here. We need the immune cells to be highly infiltrating the tumor. So the tumor has a lot of immune cells, but I suspect that they're mostly residential cell, resident cells that already trained by the tumor just to comply with the tumor biology and basically helping the tumor while the peripheral cells that can be trained to recognize the tumor antigens just don't properly infiltrate into this tissue, and so selectin is being looked at. There are a lot of publications that say E-selectin may be driving cancer, so let's knock it out. But I suspect that what is reported is the peripheral E-selectin level, not the localized tumor, localized E-selectin level. So what we need is an agonist, but we need that agonist to be specifically expressed in the tumor. I hope I didn't, didn't confuse you too much.

Brian McCloskey: Rick, do you have any other follow up questions with that?

Rick Davis: Is it possible to supplement the selectin? Because I heard you mention right at the end that you're looking at maybe selecting agonists. Is it possible, and if you were to create a trial, would you, could you include the selectin? Would it be worthwhile in the trial, including this agonist?

Ida Deichaite: That's a real good question, Rick. I think about it every day. I know that, for example, people use TLR agonists because TLR agonists help this process. But what I'm focusing on in this case is, I'm saying that TNF is a direct activator of E-selectin. It's really TNF up, selectin is up. But that's if TNF does its inflammatory act, and I'm saying that in the prostate cancer, TNF ends up being immunosuppressive. So what if I take it out and restart the system, and hopefully now I turn the TNF back to its inflammatory role, which should increase selectin expression in the tissue. So that really is the kind of idea is to reprogram TNF from immunosuppressive into an inflammatory effect, because TNF directly activates E-selectin. It's a straight line.

Rick Davis: So either this actually hits on your response to Brad and my other question in the chat, which is really confusing me, which is this: contradiction in TNF. I'm thinking, “Ah, TNF attracts the T cells.” So that's great, but at the same time, TNF seems to sustain the cold environment. So I'm a little lost on this. I'm not a scientist, as you can hear, but it's confusing me.

Ida Deichaite: I dug into the literature because it was extremely confusing to me. When I saw this connection between TNF and IL6, trust me, I didn't sleep nights. I was just like, “What is happening?” This is a real puzzle. It turns out that a lot of people looked at that. The issue is, you know, experimental systems, you have to have the right experimental systems, and mice don't have, you know, often kind of mislead us or don't are not a high fidelity representation of what happens in humans. But, you know, the cell is such that it kind of repurposes a lot of the same proteins. It just uses them in different ways based on the context. And because we only have like 20,000 genes, you know, you cannot run a body with 20,000 genes. So we end up using these genes and the proteins that they make in ways that are modular, so they can be activating or they can be suppressive. We understand that very well. It turns out that once you have an inflammation, you have to stop it at some point, because the inflammation really is a killing effect. Just kill, kill, kill. And we know that, you know, Brad just told us about how difficult it was to have this inflammatory response. So at some point, you have to stop it. And it turns out that you use similar molecules to stop it, so it's all context related. One of the surrogates of TNF being an immunosuppressive is low consistent expression. So during inflammatory response, TNF shoots up in the sky and then it disappears. But in the immune suppressive environment, it just keeps being expressed, but at very low levels, so sustained expression of TNF. It's just a surrogate. That's one of the patterns that were described in the literature. What's really interesting in clinic is that we see that in some patients who receive TNF to down regulate their autoimmune response. So you have a lot of TNF, you get autoimmune response. The TNF stays there, you knock it out, and you get rid of autoimmune response, such as Crohn's disease, colitis, psoriasis. But in some patients you actually use TNF inhibitors to treat Crohn's disease, and these patients come up with psoriasis all of a sudden. It's like they get another autoimmune disease. And one of the explanations is that we also knock out the immunosuppressive effect of TNF.

So we kill the TNF inflammatory effect, but eventually we also kill the immunosuppressive effect, and then you get the inflammation back. So that's one of the explanations why you know this

effect is observed, and one of the arguments the TNF has an immunosuppressive role in humans, and that's what I find from my studies, that in prostate cancer, the immunosuppressive environment hijacks the TNF immunosuppressive effect. Because we have TNF drugs, we can actually start looking into that, and because they have a good safety profile, and my goal is to find a way to do a clinical trial, and we could do a readout, we could do a very quick trial, which would be, you know, apply, just apply this combination of drugs and then looking at the biopsy meta analysis to see if indeed we made a difference.

Brian McCloskey: There's a question in the chat, actually one that I had too: Alexander Lalov asked, “What stage, how advanced in prostate cancer, should the patient be?” Or just to state it another way, “Are heavily pre-treated cancer patients, prostate cancer patients…would they respond better to a TNF treatment arrangement versus those that are less pre-treated?”

Ida Deichaite: Yeah, so obviously I'm not qualified to answer that question. I'm more thinking in terms of how we can address that question, how we can get to a point of it? So therefore, sort of suggest a new adjuvant approach, because once we remove the tissue, if we have metastatic tissue, it's more difficult to assess, as far as may have an effect. I couldn't say. I don't have any data to suggest that, but I'm thinking that then we still have a prostate biopsy tissue. We could pre-treat the patient with safe drugs, and then read out after we do the biopsy to see if there is a change in this particular immune cell. Or did we in some way address the immunosuppressive environment? Because we know what, what is the profile of an immunosuppressive environment? We know of T cells, B cells, etc, and also no proteins that are involved. So we could definitely have a look at that.

Brian McCloskey: Even for the 40, I think it was 49 patients that you had in the study, in the cohort, was there any stratification of patients based upon their EMR data? Perfectly pre treated, versus not?

Ida Deichaite: That's a real good question. You know what's surprising about the data? We also reproduced it in 169 patients. We looked at 49 and then we were able to identify another cohort of patients, and we did the same analysis, and it was highly reproducible. And what's interesting about it, that irrespective of the treatment, the consistency of these findings was amazing. See how consistent it is, so that's kind of interesting. As far as we can see the change in this particular expression levels of TNF and the proteins with disease progression. So we do see a change, definitely, but it's much more difficult to interpret.

Saed Sayad: Is your analysis based on just one data set you got from the other paper?

Ida Deichaite: Now we have several data sets. So as I said, the first data set was 49 patients, and then we had another three cohorts, and the data was reproducible. So 49 was a training cohort, and then 169 was a validation cohort. We haven't published it yet. We will.

Saed Sayad: But your publication is just based on one paper, one data set?

Ida Deichaite: Yes, but yes, like I said, we're in the process of writing our second paper, which will be interesting, because we are going to look at IRC immunohistochemistry, identifying different immune cells in the prostate tissue as compared to BPH and stroma. So that will be our next paper, and indeed, then we looked at 169 patients. All data was irreducible.

Brad Power: One of Brian's favorite questions is, “How could diagnostics and tests around these dynamics point to patients that would be more or less likely to respond?” I'm thinking in particular, that BostonGene is very good at multiplex immunofluorescence and spatial and proteomics and these kinds of analysis, of not just the tumor cell, but the environment around the tumor cell. I’m curious, if there's anything that is on the horizon, or you might do. I know you worked, you said, with Caris, but just trying to get at the surround of diagnostics that might personalize, in some ways, the treatment.

Ida Deichaite: Well, Brad, this is like the best question, because my dream, if I could say what I would really like is to see a large cohort of data of immune response, of checkpoint response, in relation to E-selectin, in relation to IL6, 1b, etc, TNF. And the data set I have has to do with Caris. We're dealing with 90 patients, and when we look at the quartiles of expression, we just lose any statistical power there. So if BostonGene has more data on patients with checkpoint inhibitors, we could correlate that data with the biomarkers we identify, that would be just a dream come true.

Brian McCloskey: Michael Hensley from BostonGene reached out to me while you were talking, and wants an introduction. So, Ida, this could present a great opportunity for you, because they've got nothing but data.

There's a question from Stephen Barry: would you combine TNF alpha with androgen ablation blockers in your trial? And your trial would be only for primary cancer, neoadjuvant setting.

Ida Deichaite: When it comes to AR (androgen receptor), it's very complicated. There is some data suggesting that TNF and AR have some kind of interconnection, or there is some kind of connection there. I haven't looked into that. I couldn't answer this question, just my understanding of how these two work together.

Brian McCloskey: There was a question from Alane Watkins regarding CTCs (circulating tumor cells). Have you found any relationship between CTCs and TNF?

Ida Deichaite: Sorry. Alane, no. We didn't have a chance to look at that. It's a good question though.

Brian McCloskey: You said at the outset that this was going to be a patient-oriented one, but I think we got into quite a bit of science today, which was great. The great thing is that our group here can go into the science, which is awesome. We have a really smart group of people here. Ida, thank you so much for your time, for educating us.

“How a Particular Protein (Tumor Necrosis Factor) Might Control the Immune Response in Prostate Cancer” (Ida Deichaite, PhD) [#118]

CHAT DISCUSSION

00:16:53 Robb Owen: Are you doing research into how zinc regulates epithelial cells and protects them from apoptosis due to TNF. Zinc also lowers levels of pro-inflammatory cytokines, including TNF.
00:24:06 Robb Owen: Have you considered how cimetidine, as a Histamine H2 inhibitor will inhibit TNF production and when used with zinc, vitamin C and other antiemetics like prochlorperazine and GABA inhibitors like Alprazolam manage neurotransmitter inhibition or regulation of the following, TNF, NF-kB, P13k/AKT, VEGF, FGF, Notch, etc?
00:35:15 Alane Watkins: Have you found any relationship between CTCs and TNF?
00:40:51 Rick Davis, AnCan Foundation: Can you supplement/increase SELE? In a clinical trial, would adding SELE to the TNF inhibitor be a second variable to test? What led you to SELE?
00:43:29 Robb Owen: Rick, yes, Selenium is critical, I ensure that patients have the correct levels of selenium due to the importance of selenoproteins in the TME and oxidative stress on DNA. Genistein is crucial as well.
00:45:16 Rick Davis, AnCan Foundation: @Robb - isn't there evidence that excess selenium is adverse to PCa?
00:46:54 Rick Davis, AnCan Foundation: Help us understand the contradiction that TNF on the one hand attracts T-cells from inflammation, but on the other hand sustains the cold tumor?
00:47:52 Alexander Lalov: What stage (how advanced) prostate cancer has to be in order to receive the most benefits...
00:48:39 David Plunkett: I am having difficulty not conflating Selectin and Selenium.
00:50:13 stephenbarry: Would you combine TNF alpha with androgen ablation/blockers in you trial? And your trial would be only for primary cancer, neoadjuvant setting?
01:03:29 John Tepe: Thanks Ida for your presentation.
01:03:45 David Plunkett: Hard stop for me, sorry.
01:04:16 Dr. Chris Apfel: Great session. Thank you!
01:04:58 Alane Watkins: Wonderful discussion today. Thank you all!
01:05:34 Ryan Moon (he/him): Thanks!